PROBLEM 15

Stokes Drag and Terminal Velocity

Original statement · Problems.pdf, p. 9

Viscous Aerodynamics

The Stokes' law states that the magnitude of the drag force acting on a sphere of radius r and velocity v moving in a fluid at rest of dynamic viscosity μ is: $F(v) = 6\pi\mu r v$. Assuming the density of a vertically-falling sphere is twice the density of the fluid (ρ_0), and considering drag, the weight, and the buoyancy of the sphere, compute:

(a) The terminal velocity of the sphere.

(b) The evolution of the position as a function of time, if the sphere starts from rest.

Remember that the kinematic viscosity is defined as $\nu = \mu/\rho_0$.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A solid sphere of radius r and density ρ_s falls vertically from rest through a quiescent fluid of density ρ_0 and dynamic viscosity μ.
- **Forces Acting on the Sphere:**

1. Gravity (Weight): $W = mg = \rho_s V g$, acting downwards.

2. Buoyancy (Archimedes' Principle): $F_B = \rho_0 V g$, acting upwards.

3. Stokes Viscous Drag: $F_D = 6\pi\mu r v$, acting upwards (opposing downward motion $v > 0$).

- **Geometric & Material Relations:**
- Sphere volume: $V = \frac{4}{3}\pi r^3$.
- Given sphere density: $\rho_s = 2\rho_0$.
- Sphere mass: $m = \rho_s V = 2\rho_0 \left(\frac{4}{3}\pi r^3\right) = \frac{8}{3}\pi\rho_0 r^3$.
- Fluid kinematic viscosity: $\nu = \frac{\mu}{\rho_0} \implies \mu = \rho_0 \nu$.
- **Initial Conditions:**

$$x(0) = 0, \qquad v(0) = \dot{x}(0) = 0$$

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Frame:** Fixed Cartesian frame with the vertical axis Ox directed **downwards** along gravity.
- **Resultant Force:**

$$\sum F_x = W - F_B - F_D$$

Substitute the expressions for each force:

$$W = \left(\frac{8}{3}\pi\rho_0 r^3\right) g$$

$$F_B = \rho_0 \left(\frac{4}{3}\pi r^3\right) g = \frac{4}{3}\pi\rho_0 r^3 g = \frac{1}{2}W$$

$$F_D = 6\pi\mu r\dot{x} = 6\pi(\rho_0\nu)r\dot{x}$$

Net buoyancy-corrected gravitational force:

$$W - F_B = \frac{8}{3}\pi\rho_0 r^3 g - \frac{4}{3}\pi\rho_0 r^3 g = \frac{4}{3}\pi\rho_0 r^3 g = \frac{1}{2}mg$$

PHASE 3

Step-by-Step Mathematical Deduction

1. Derivation of the Equation of Motion

Pedagogical Justification: Applying Newton's second law $\sum F_x = m\ddot{x}$ establishes a linear first-order ODE in velocity $v = \dot{x}$.

$$m\ddot{x} = (W - F_B) - F_D$$

$$\left(\frac{8}{3}\pi\rho_0 r^3\right)\ddot{x} = \frac{4}{3}\pi\rho_0 r^3 g - 6\pi\rho_0\nu r\dot{x}$$

Divide both sides by $\frac{8}{3}\pi\rho_0 r^3$:

$$\ddot{x} = \frac{\frac{4}{3}\pi\rho_0 r^3 g}{\frac{8}{3}\pi\rho_0 r^3} - \frac{6\pi\rho_0\nu r}{\frac{8}{3}\pi\rho_0 r^3}\,\dot{x}$$

Simplify each fraction:

$$\frac{\frac{4}{3}}{\frac{8}{3}} = \frac{1}{2} \implies \text{First term: } \frac{1}{2}g$$

$$\frac{6\pi\rho_0\nu r}{\frac{8}{3}\pi\rho_0 r^3} = 6 \cdot \frac{3}{8}\,\frac{\nu}{r^2} = \frac{18}{8}\,\frac{\nu}{r^2} = \frac{9\nu}{4r^2}$$

Therefore, the differential equation reads:

$$\ddot{x} + \frac{9\nu}{4r^2}\,\dot{x} = \frac{1}{2}g$$

Multiplying through by $4r^2$:

$$4r^2\ddot{x} + 9\nu\dot{x} = 2r^2 g$$

Compare with the official equation of motion in the official solution key (Problems.pdf, Solution keys section).

2. Computation of Terminal Velocity v_t

> **Pedagogical Justification:** The terminal velocity is reached when the viscous drag balances the effective weight, so acceleration ceases ($\ddot{x} = 0$).

$$4r^2(0) + 9\nu v_t = 2r^2 g \implies v_t = \frac{2r^2 g}{9\nu}$$

3. Analytical Integration for Velocity $v(t)$

Let $\gamma = \frac{9\nu}{4r^2}$ be the characteristic inverse relaxation time ($[\gamma] = \mathrm{s}^{-1}$).

The ODE becomes:

$$\dot{v} + \gamma v = \frac{1}{2}g = \gamma v_t$$

$$\frac{dv}{dt} = -\gamma(v - v_t) \implies \frac{dv}{v - v_t} = -\gamma\, dt$$

Integrate with initial condition $v(0) = 0$:

$$\int_0^{v(t)} \frac{dv'}{v' - v_t} = -\gamma \int_0^t dt'$$

$$\left[\ln|v' - v_t|\right]_0^{v(t)} = -\gamma t$$

$$\ln\left(\frac{v_t - v(t)}{v_t}\right) = -\gamma t$$

Exponentiate both sides:

$$1 - \frac{v(t)}{v_t} = e^{-\gamma t} \implies v(t) = v_t\left(1 - e^{-\gamma t}\right)$$

4. Analytical Integration for Position $x(t)$

> **Pedagogical Justification:** Position is the direct integral of velocity over time, $x(t) = \int_0^t v(t')\,dt'$ with $x(0) = 0$.

$$x(t) = \int_0^t v_t \left(1 - e^{-\gamma t'}\right) dt' = v_t \left[t' + \frac{1}{\gamma} e^{-\gamma t'}\right]_0^t$$

Evaluate at upper and lower limits:

$$x(t) = v_t \left(t + \frac{1}{\gamma} e^{-\gamma t} - \left(0 + \frac{1}{\gamma}\right)\right) = v_t \left(t - \frac{1}{\gamma} + \frac{1}{\gamma} e^{-\gamma t}\right)$$

Factor out $\frac{1}{\gamma}$:

$$x(t) = \frac{v_t}{\gamma} \left[\gamma t - 1 + e^{-\gamma t}\right]$$

Now substitute $v_t = \frac{2r^2 g}{9\nu}$ and $\gamma = \frac{9\nu}{4r^2}$:

$$\frac{v_t}{\gamma} = \frac{\frac{2r^2 g}{9\nu}}{\frac{9\nu}{4r^2}} = \frac{2r^2 g}{9\nu} \cdot \frac{4r^2}{9\nu} = \frac{8r^4 g}{81\nu^2}$$

Rewrite $\frac{v_t}{\gamma}[\gamma t - 1 + e^{-\gamma t}]$:

$$x(t) = \frac{8r^4 g}{81\nu^2} \left[\frac{9\nu}{4r^2} t - 1 + e^{-\frac{9\nu}{4r^2} t}\right]$$

Multiply the inside by $4r^2$ and divide the outside prefactor by $4r^2$:

$$\frac{8r^4 g}{81\nu^2} = \frac{2r^2 g}{81\nu^2} \cdot 4r^2$$

$$x(t) = \frac{2r^2 g}{81\nu^2}\left[4r^2\left(\frac{9\nu}{4r^2}t\right) - 4r^2 + 4r^2 e^{-\frac{9\nu}{4r^2}t}\right]$$

$$x(t) = \frac{2r^2 g}{81\nu^2}\left[-4r^2 + 4r^2\exp\left(-\frac{9\nu}{4r^2}t\right) + 9\nu t\right]$$

Compare with the closed-form solution in the official solution key (Problems.pdf, Solution keys section).

PHASE 4

Physical Interpretation & Dimensional Verification

- **Short Times ($t \ll \gamma^{-1}$):**

Expand the exponential $e^{-\gamma t} \approx 1 - \gamma t + \frac{1}{2}\gamma^2 t^2$:

$$x(t) = \frac{v_t}{\gamma}\left[\gamma t - 1 + \left(1 - \gamma t + \frac{1}{2}\gamma^2 t^2\right)\right] = \frac{v_t}{\gamma}\left(\frac{1}{2}\gamma^2 t^2\right) = \frac{1}{2}(\gamma v_t)t^2 = \frac{1}{2}\left(\frac{1}{2}g\right)t^2 = \frac{1}{4}gt^2$$

This corresponds to free fall under effective gravity $g_{eff} = g(1 - \rho_0/\rho_s) = g/2$.

- **Long Times ($t \gg \gamma^{-1}$):**

The exponential term decays to zero:

$$x(t) \approx v_t t - \frac{v_t}{\gamma}$$

The sphere reaches uniform terminal descent at speed v_t with an apparent positional offset $\Delta x = v_t / \gamma$.

- **Dimensional Verification:**

$[v_t] = [r^2 g / \nu] = \frac{\mathrm{m}^2 \cdot \mathrm{m} \cdot \mathrm{s}^{-2}}{\mathrm{m}^2 \cdot \mathrm{s}^{-1}} = \mathrm{m/s}$. Correct.

Source: Problems.pdf